DataTürk Project

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***Abstract*—This document is the first progress report for DataTürk's machine learning project. The document contains the details of the dataset cleansing process and the summary of the given dataset.**

***Index Terms*—machine learning, github, dataset, missing values, graphs**

I. Introduction

We have created a python script [1], in order to identify the missing values in our given dataset, and analyze the data according to their contextual meaning.

II. Data Preprocessing

A. Dropping Some of the Columns and Filling the Values

We identified the columns that contained more than 50 percent of the missing data values in them and filled in these numeric columns using the calculated mean of such columns [2]. For non-numeric columns, we used the mode. In order to eliminate the bugs in the code, we took advantage of some sources [3] [4] and ChatGPT-3.

B. Creating a Binary Label for the Dataset

We converted the last columnt to a binary label column, which gave us a way to tag and identify these columns.

III. Appendices

A. Summary of the Dataset

Missing values	0
Non-numerical columns	13
Numerical columns	68
Rows	246008
Columns	81

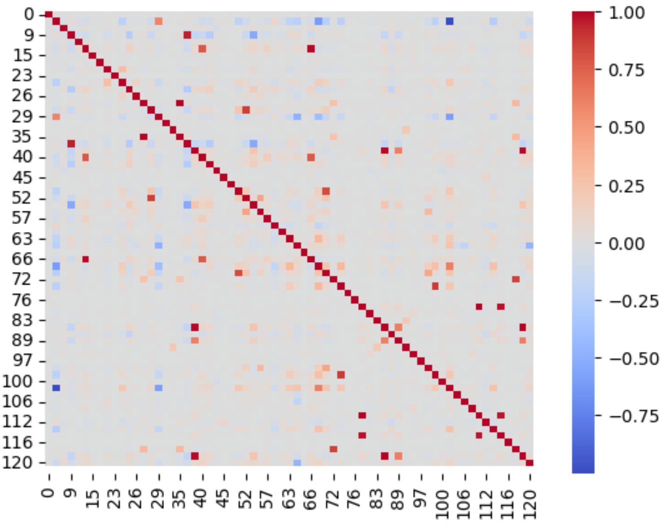

Fig. 1. Correlation between information on the dataset.

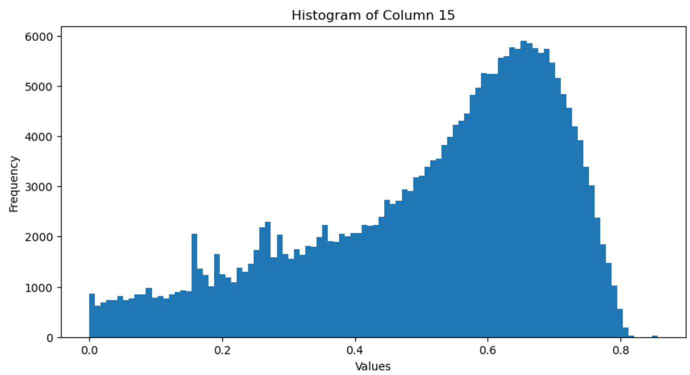

Fig. 2. Histogram of Column 15.

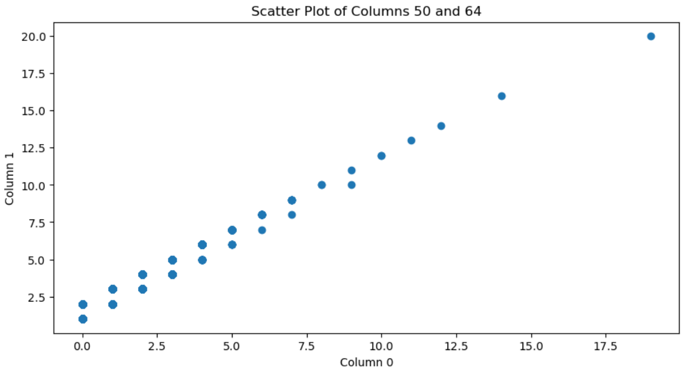

Fig. 3. Positive correlation between the column 50 and 64.

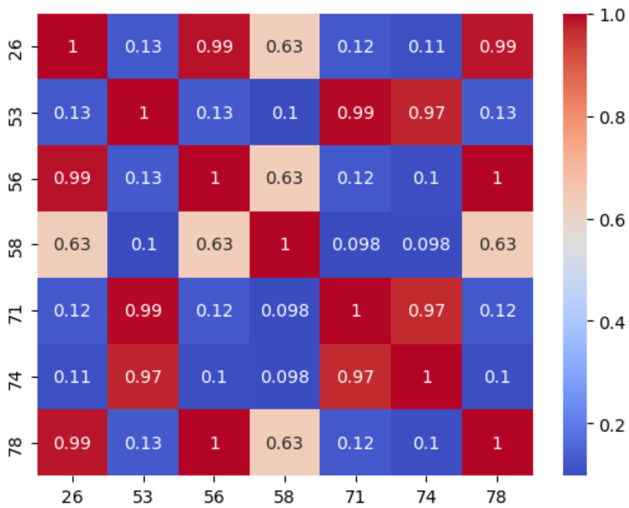

Fig. 4. .

IV. Conclusion and Future Work

Our next step will be to go on the depths of machine learning with our cleansed dataset.

References

[1] iamemirsahin. DataTürk. [Online]. Available: https://github.com/iamsahinemir/DataTurk
[2] Cook, A. Handling Missing Values. [Online]. Available: https://www.kaggle.com/code/alexisbcook/handling-missing-values
[3] Güneş, A. Python ile Veri Temizleme ve Ön İşleme. [Online]. Available: https://alprgunes.medium.com/
[4] TirendazAkademi. Nasıl Veri Bilimcisi Olunur. [Online]. Available: https://tirendazakademi.medium.com/